

VASANTH B

DATA ENGINEER

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PROFESSIONAL SUMMARY

Data Engineer with 3+ years of experience designing, building, and optimizing scalable ETL/ELT pipelines on AWS for the Banking and Healthcare domains. Proficient in PySpark, Apache Spark, and Apache Airflow, with deep expertise in Lakehouse Architecture and Dimensional Modelling. Experienced in transitioning traditional batch processing workloads to near real-time data pipelines, improving query performance by over 80%, and implementing robust data quality frameworks. Adept at leveraging Databricks, dbt, and Amazon Redshift to transform raw, high-volume data into trusted, analytics-ready datasets that drive critical business decisions.

TECHNICAL SKILLS

Cloud & Data Platforms: AWS (S3, Glue, EMR, Redshift, Kinesis, Athena, Lambda), Databricks, Delta Lake, Lakehouse Architecture (Medallion: Bronze, Silver, Gold)

Programming & Querying: Python (PySpark, Pandas, PyTest), SQL (CTEs, Window Functions, Stored Procedures, Query Optimization), Bash/Shell Scripting

Data Engineering & Big Data: Apache Spark, Spark SQL, ETL/ELT Pipeline Development, Batch & Streaming Processing, Change Data Capture (CDC), Data Ingestion, Data Partitioning

Data Modelling & Warehousing: Dimensional Modeling, Star Schema, Slowly Changing Dimensions (SCD Type 1 & 2), dbt, Query Performance Tuning

Databases: PostgreSQL, MySQL, Microsoft SQL Server

Workflow Orchestration: Apache Airflow, AWS Step Functions

Data Quality & Monitoring: Great Expectations, Data Profiling, Data Validation, Monitoring & Logging, Amazon CloudWatch

DevOps & CI/CD: Git, GitHub, Docker, GitHub Actions, AWS CodePipeline

Analytics & Visualization: Power BI (DAX), Jupyter Notebook

Data Formats & Methodologies: Parquet, ORC, Avro, Agile/Scrum

PROFESSIONAL EXPERIENCE

Data Engineer | [Globe Infotech Pvt.Ltd](#)

Oct 2025 – Present

Project: Banking Transaction Data Platform & Healthcare Claims Pipeline Modernization

- Designed and developed scalable ETL/ELT pipelines for banking and healthcare domains, processing 500GB–1TB of data daily. Leveraged PySpark, AWS Glue, and Databricks to handle large-scale data transformations, ensuring high throughput and fault tolerance across distributed computing environments.
- Implemented near real-time data ingestion using Amazon Kinesis and Change Data Capture (CDC) mechanisms to capture operational changes instantly. Reduced batch processing latency by 30% through careful Catalyst optimizer tuning and Parquet/ORC partition pruning, optimizing data reads across hundreds of millions of records.
- Built a Medallion Architecture (Bronze, Silver, Gold) leveraging Databricks and Delta Lake ACID transactions on Amazon S3. Standardized data quality and schema enforcement at each layer, exposing curated Gold-layer tables via Amazon Athena to maintain sub-minute data freshness SLAs for downstream analytics.
- Orchestrated complex data workflows using Apache Airflow, scheduling DAGs with robust task dependencies, retries, and SLA monitoring. Integrated pipelines with AWS Lambda and Step Functions for event-driven triggers, sustaining reliable execution and high availability during peak transaction spikes.
- Developed reusable dbt transformations featuring incremental loads to optimize compute costs and processing time. Integrated automated data validation and monitoring via Amazon CloudWatch to proactively catch anomalies, improving overall data reliability and reducing downstream reporting errors by 40%.
- Established automated CI/CD pipelines using GitHub Actions and Docker, containerizing applications for consistent environments. Automated testing and deployment workflows, reducing release time from days to 15 minutes and enabling frequent, reliable production updates.

Associate Data Engineer | [Troika Research Technologies](#)

Jan 2024 – Sep 2025

Project: Customer Transaction Data Processing System / Banking / Financial Services

- Developed serverless ETL pipelines to process daily banking transactions using PySpark, AWS Glue, and Pandas. Efficiently ingested data from heterogeneous sources including PostgreSQL, MySQL, Microsoft SQL Server, and REST APIs via AWS Lambda and AWS Step Functions.
- Developed and scheduled Apache Airflow DAGs to automate daily batch ETL workflows. Implemented robust task retry mechanisms, dependency management, and automated failure alerting to ensure reliable pipeline execution and timely data delivery to stakeholders.
- Built multi-zone Data Lake structures (Raw, Processed, Curated) on Amazon S3 using Delta Lake ACID transactions and S3 Intelligent-Tiering. Ensured sub-minute data availability for downstream consumers while reducing overall storage costs by 35% through effective data lifecycle management.
- Authored complex SQL transformations using CTEs, Window Functions, and Stored Procedures to model intricate financial data relationships. Implemented dbt data quality checks featuring automated schema drift detection, improving financial reconciliation accuracy by 25%.
- Optimized Amazon Redshift cluster performance by analyzing query execution plans and configuring distribution keys, sort keys, and WLM queue configurations. Reduced average BI query latency from 45s to 8s for heavy enterprise workloads, and supported downstream Power BI reporting through optimized dimensional models and DAX measures.

- Developed ETL workflows to extract, clean, and transform customer marketing campaign data from multiple structured sources into analytics-ready datasets.
- Applied Python preprocessing using Pandas and NumPy to resolve missing values and duplicate records, improving dataset accuracy before loading.
- Wrote SQL queries using GROUP BY, CTEs, subqueries, and window functions to analyze campaign performance and customer behavior trends.
- Loaded transformed datasets into Amazon Redshift, contributing to a centralized warehouse used for reporting and analytics across teams.
- Performed schema validation and consistency checks to strengthen data reliability for downstream marketing analytics use cases.
- Prepared analytics-ready datasets that supported a customer response prediction model for targeted marketing analysis.
- Built Power BI dashboard components to visualize campaign performance and customer response metrics.

EDUCATION

Bachelor of Engineering (Computer Science and Engineering)

Nehru Institute of Engineering and Technology, Coimbatore

May 2023